

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	SWTID-2026-4993
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Homepage UI	Attractive landing page with project introduction	UI	Done	High
2	Crop prediction	Takes user input for soil/environment values	Frontend	Done	High
3	Input Validation	Validates input fields before prediction	Frontend + Backend	Done	Medium
4	ML Prediction engine	Predict best crop using trained model	Backend / ML	Done	High
5	Data Preprocessing	Uses scaler for preprocessing input values	ML pipeline	Done	High
6	Result Design	Shows recommended crop and confidence score	UI + Backend	Done	High
7	Responsive Design	Works on desktop and mobile devices	UI	Done	Medium
8	Error Handling	Handles invalid inputs and runtime errors	Backend	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Homepage, Form, Result Page
2	Backend / Logic	3	Prediction logic, Validation, Error Handling
3	Database / Storage	0	Not Used
4	API / Integration	1	Flask Prediction Route
5	Security / Authentication	0	Not Implemented